

# The atom and its particles

## What is an atom?

Atoms are the building blocks that make up all the matter around us: the ground, the air and all gases, water and all liquids, food, chemical elements, and even our own body. Everything around us, including ourselves, is made of atoms.

! Very important

The atom is the smallest unit of matter that retains the properties of an element. All the atoms known to humanity are organized in the [Periodic Table](#).

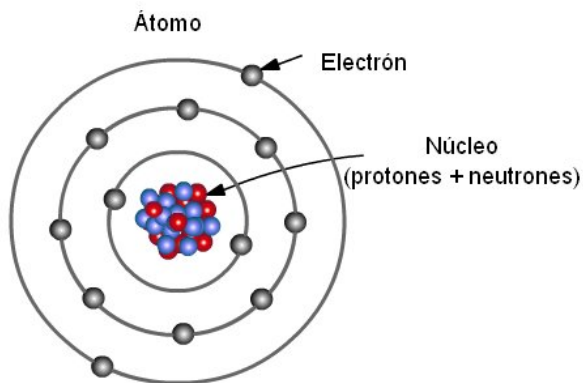
Atoms are extremely small (nanometric scale, between  $0,1 \cdot 10^{-9}m$  and  $0,5 \cdot 10^{-9}m$ ): they cannot be seen with the naked eye or with an optical microscope.

i Learn more

Atoms can only be seen with special microscopes such as: [Scanning Tunneling Microscope \(STM\)](#), [Atomic Force Microscope \(AFM\)](#), [High-resolution Transmission Electron Microscope \(TEM/STEM\)](#).

## Subatomic particles

Here we show a model of the atom based on the model proposed by [Bohr](#). We say it is a model because the atom is actually much stranger. According to this model, the atom can be drawn like this:



Atoms are made up of three main particles:

| Particle | Symbol in notes | Electric charge | Mass (approx.) | Location                                                |
|----------|-----------------|-----------------|----------------|---------------------------------------------------------|
| Proton   | $p^+$           | +1              | 1              | Nucleus                                                 |
| Neutron  | $n$             | 0               | 1              | Nucleus                                                 |
| Electron | $e^-$           | -1              | $\sim 0$       | Orbit around the nucleus in the shell (electron cloud). |

- Protons and neutrons are in the nucleus.

- Electrons move around the nucleus in orbitals like planets orbiting the Sun.
- Neutrons and protons are much heavier than electrons. Therefore, when calculating mass, we only add protons and neutrons.

## Very important numbers

### Atomic number (Z)

The **atomic number (Z)** is the number of protons in an atom.

- It defines the element.
- Example: Hydrogen has  $Z = 1$ . This can be seen in the periodic table.

$$Z = n^{\circ} p^{+}$$

Where  $n^{\circ} p^{+}$  = number of protons.

### Mass number (A)

The **mass number (A)** is the sum of the number of protons and neutrons. It tells us the mass of a specific atom (remember that electrons are so light that we ignore them for mass).

$$A = Z + n n$$

Where:

- $Z$  = number of protons.
- $n n$  = number of neutrons.

### Electric charge (Q)

To calculate the electric charge, you only need to subtract the number of electrons from the number of protons.

$$Q = n p^{+} + n e^{-}$$

If  $Q$  is 0, we say the atom is neutral, which means it has no charge.

If  $Q$  is positive, the atom has a positive charge.

If  $Q$  is negative, the atom has a negative charge.

 Learn more

Video explaining [electric charge](#)

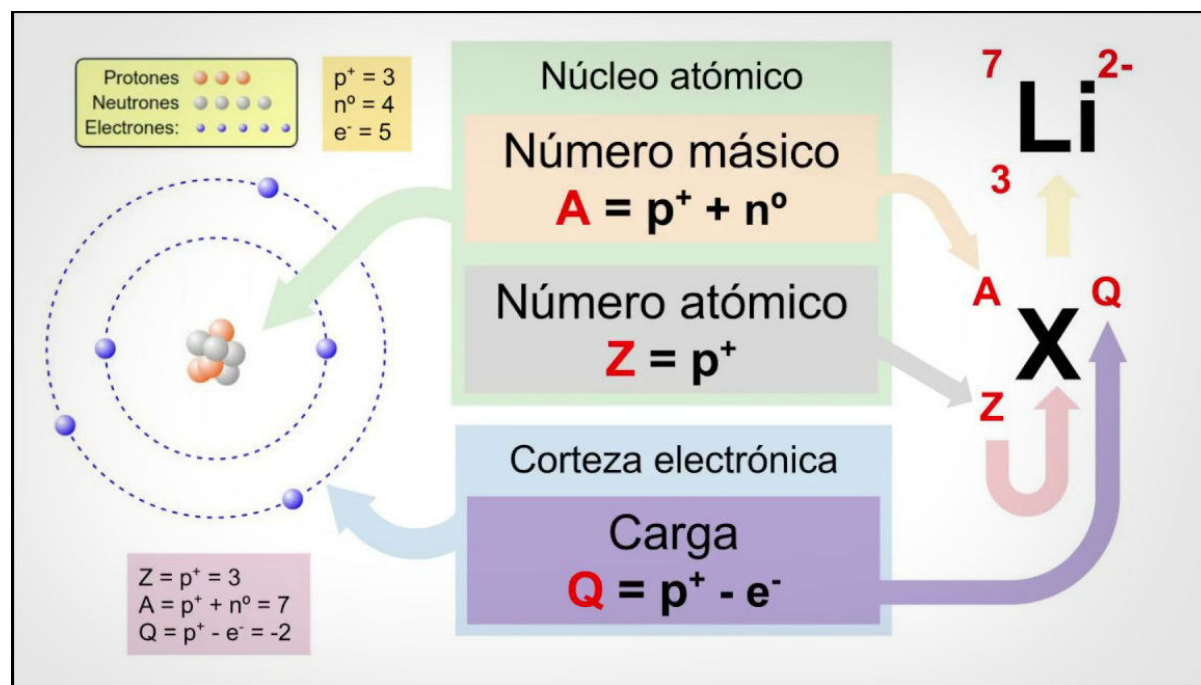
 Very important

Summary of important numbers:

1. Atomic number:  $Z = n p^{+}$
2. Mass number:  $A = Z + n n$

3. Electric charge:  $Q = n p^+ + n e^-$

With these numbers, we usually identify an atom in a very simplified way like this:



## Isotopes

Isotopes are atoms of the same element with:

- The same number of protons ( $Z$ )
- A different number of neutrons ( $n$ )

An example is carbon isotopes: Carbon-12 and Carbon-14. They are written as: *element-A*.

Example of hydrogen isotopes:

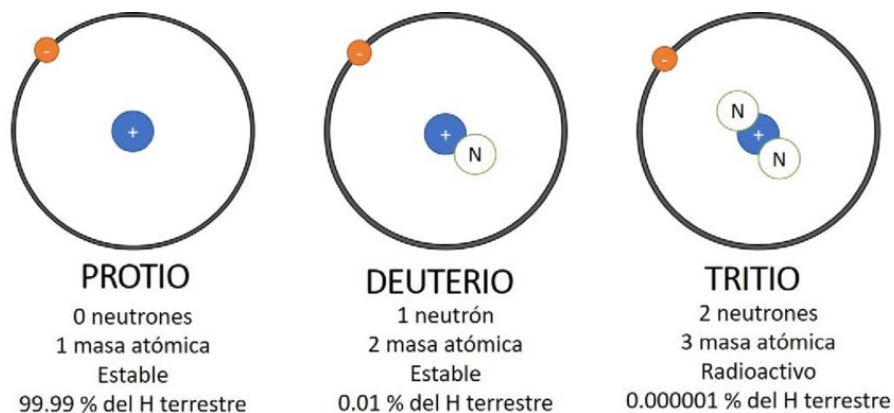


Figure 1: Hydrogen isotopes

# Ions

Ions are atoms that have an electric charge, meaning the number of protons is different from the number of electrons. Let's break it down:

- A **neutral atom** is the usual type of atom. This means its electric charge is zero. This happens when we have the same number of protons ( $p^+$ ) and electrons ( $e^-$ ). Each positive charge of a proton is balanced by a negative charge of an electron, making the total charge zero.
- An atom that does have an electric charge (that is:  $Q \neq 0$ ) is called an **ion**. Remember this happens when: protons  $\neq$  electrons

Types of ions:

- **Cation**: loses electrons  $\rightarrow$  positive charge ( $Q > 0$ ).
- **Anion**: gains electrons  $\rightarrow$  negative charge ( $Q < 0$ ).

## Relationship with the periodic table

- The periodic table is ordered by atomic number ( $Z$ ).
  - Each element has a unique  $Z$ .
  - Elements in the same group have similar properties.
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## Summary

### ! SUPER SUMMARY

- Atoms are made up of protons (+) and neutrons in the nucleus, and electrons (-) in the electron cloud.
  - $Z$  = number of protons.
  - $A$  = protons + neutrons.
  - $Q$  = protons - electrons. Ions have charge:
    - Cation: atom with  $Q > 0$
    - Anion: atom with  $Q < 0$
  - Isotopes differ in the number of neutrons.
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## Exercises

Complete the following table:

| Element | Z  | A  | Protons | Neutrons | Electrons |
|---------|----|----|---------|----------|-----------|
| X       | 8  | 16 | ?       | ?        | ?         |
| Y       | ?  | 23 | 11      | ?        | 11        |
| Z       | 17 | ?  | ?       | 18       | ?         |

Based on the table above:

1. Calculate the missing values.
2. Indicate whether the atom is neutral or an ion.
3. If it is an ion, indicate its charge.

## Challenge question

An atom has: - 12 protons - 12 neutrons - 10 electrons

Answer:

- What is Z?
- What is A?
- Is it an ion?
- What is its charge?